EXPRIMENT-14

14.) One-Time Pad Version of Vigenère Cipher

Aim:

To write a Python program to encrypt and decrypt a message using the one-time pad version of the Vigenère cipher.

PROGRAM:



```
def encrypt(text, key):
    result = ""

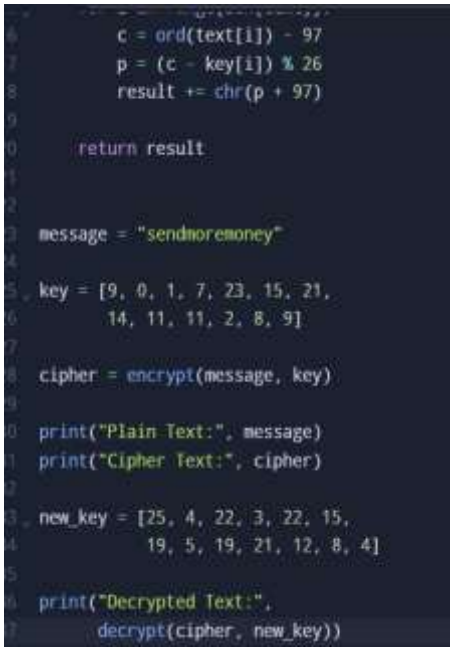
    for i in range(len(text)):
        p = ord(text[i]) - 97
        c = (p + key[i]) % 26
        result += chr(c + 97)

    return result


def decrypt(text, key):
    result = ""

    for i in range(len(text)):
        c = ord(text[i]) - 97
        p = (c - key[i]) % 26
        result += chr(p + 97)

    return result
```



```
        c = ord(text[i]) - 97
        p = (c - key[i]) % 26
        result += chr(p + 97)

    return result


message = "sendmoremoney"

key = [9, 0, 1, 7, 23, 15, 21,
       14, 11, 11, 2, 8, 9]

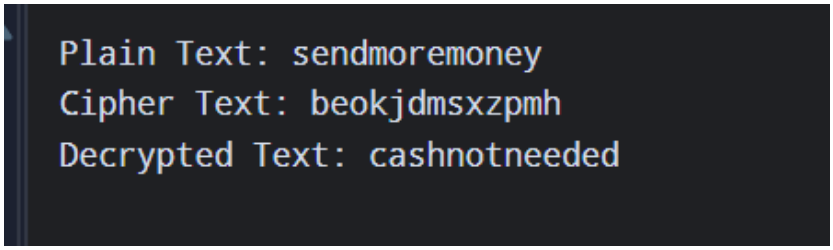
cipher = encrypt(message, key)

print("Plain Text:", message)
print("Cipher Text:", cipher)

new_key = [25, 4, 22, 3, 22, 15,
           19, 5, 19, 21, 12, 8, 4]

print("Decrypted Text:",
      decrypt(cipher, new_key))
```

OUTPUT:



```
Plain Text: sendmoremoney
Cipher Text: beokjdmsxzpmh
Decrypted Text: cashnotneeded
```

RESULT:

Thus, the message was successfully encrypted and decrypted using the one-time pad Vigenère cipher.